DESIGN I

SOFTWARE ENGINEERING II

TESTIGOS DE MELUK

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System domain

System context

Industrias Escolares is a school furniture manufacturing company (chairs, tables, and desks) with a workforce of 10 to 20 employees. In its current setup, the owner provides quotes directly and generates the work order once the request is ready, which is subsequently invoiced. This entire process is recorded in the desktop version of Siigo Pyme, installed on a single, non-shared computer. This dynamic defines the central restriction of the new design: the proposed system must coexist with Siigo without replacing it or writing to it automatically, as there is no public API and tax accounting will remain the exclusive responsibility of the accounting software.

The actual operational flow that the new system must support remains intact: the owner provides quotes, generates the work order, and the secretary logs it into Siigo. The integration of a virtual assistant will solely serve to feed the initial phase with preliminary quotes, without replacing anyone in the current workflow. A key factor at this stage is the absence of an official, up-to-date price list since January 2016. The final price depends primarily on customer tenure and order volume, variables that are adjusted based on workshop workload, raw material costs, material availability, and transportation. For this reason, the system will suggest price ranges rather than imposing fixed rates.

Finally, the physical environment heavily dictates the security design. The computer hosting Siigo is located facing the door, leaving information exposed. Data backup is performed manually on a single USB drive that only management connects at closing time, and there is a Mac computer where company and personal files are mixed across two USB drives. Add to this that the network runs exclusively on Wi-Fi with no guaranteed hardwiring, and the total hardware inventory is limited to two desktop computers and one laptop. This highlights that the security problem transcends the digital realm and includes physical vulnerabilities; consequently, the new system's security module must combine strict account and session control with a verifiable backup discipline.

Characterization of actors

Actor	Type	Characterization
Owner / Management	Internal user, Administrator role	Provides quotes directly without intermediaries, generates work orders (W.O.), approves or rejects quotes generated by the assistant, and determines final prices with proper justification. She is the only person authorized to connect the backup USB drive and manage the DIAN (tax authority) password, although the assistant types it in. Additionally, she makes all decisions regarding production schedule saturation.

Secretary / Accounting Assistant	Internal user, Operator role	Responsible for logging the information generated by management (such as work orders) into Siigo, as well as importing data and viewing customer profiles. Operates the shared computer, which is subject to an automatic lock due to inactivity.
Accounting	Internal user, Operator role	Primarily responsible for reviewing financial reports and accessing customer profiles.
Potential Customer	External user, no account	Can request quotes 24/7 through the virtual assistant using a guided form (requiring name, email, and phone number). Receives a preliminary quote in PDF format or via email. Under no circumstances do they have access to internal pricing or system logs.
Super Administrator	System Administrator	Responsible for creating and deactivating accounts, with mandatory role assignment. Also manages the backup inventory and performs restoration tests in an isolated environment (sandbox).
Siigo Pyme	External system, non-human	The authoritative accounting software, installed on a single computer with no public API. It receives data via manual uploads from files exported by the new system. It operates as a one-way workflow, with no automatic data entry or real-time synchronization.
Backup Service	Subsystem, non-human	Executes differential backups daily at closing time. Maintains a retention scheme that includes 7 daily, 4 weekly, and 12 monthly backups. Furthermore, it performs data integrity checks, issues reminders, and manages a second physical hard drive for rotation.

Business rules

Business Rules for Module 1 (Security and Backups)

- **RN-M1-01:** All user accounts are individual and require a mandatory role assignment (Super Administrator, Administrator, or Operator). Accounts cannot be created without a defined role, and system permissions depend exclusively on this role, never on the specific individual.

- **RN-M1-02:** No session can operate without two-factor authentication. Once a valid password is provided, the system requires a six-digit temporary code before issuing access tokens. Three consecutive failed attempts will temporarily lock the account.
- **RN-M1-03:** Only one active session is allowed per account. A new login automatically invalidates the previous one. Additionally, after fifteen minutes of inactivity, the session is automatically closed on the shared computer.
- **RN-M1-04:** The system performs a daily data backup at closing time, maintaining a rotation scheme that keeps seven daily, four weekly, and twelve monthly copies. This process includes a data integrity check and triggers an alert if the USB drive is not connected. Any data restoration must first be tested in an isolated sandbox environment before being applied to production.
- **RN-M1-05:** Every critical action within the system, such as price changes, role modifications, logins, or backup executions, is recorded in an unalterable audit log. This record details who performed the action, when it occurred, and the exact reason behind it.

Business Rules for Module 2 (Commercial and Assistant)

- **RN-M2-01:** Since there is no official, up-to-date price list, the catalog displays reference prices and a suggested range based on historical sales data, using a default and configurable three-month window. If no price list is found, the catalog opens with a warning but never blocks operations.
- **RN-M2-02:** Only management is authorized to modify prices. This change must be supported by a justification document and recorded in the audit log, detailing the previous price, the new price, the date, the responsible user, and the reason. Any attempt to modify prices by another role will be denied and logged in the audit trail.
- **RN-M2-03:** The virtual assistant is restricted to generating preliminary quotes that fall within the reference price range. This process is carried out through a guided form that prevents free-text entry, and the document is sent via the WhatsApp Business API with a disclaimer stating it is preliminary and subject to approval.
- **RN-M2-04:** No quote can advance in the workflow without human approval. The owner is responsible for approving, amending, or rejecting requests, always stating the reason. If a quote is rejected, it reverts to its preliminary status and never enters the export queue for Siigo.
- **RN-M2-05:** The estimated delivery date corresponds to the dispatch date, which is calculated using the historical median based on product and volume, while adding time buffers for potential absenteeism or material shortages. If there is insufficient historical data to perform the calculation, the system clearly reports this and refrains from fabricating a date.
- **RN-M2-06:** Every work order has a unique number and follows a sequential cycle: it starts as pending, is then logged into Siigo by the secretary, and is finally marked as reconciled. The document includes mandatory fields for polypropylene, paint, wood details such as bevels, lacquered edges, description, and Formica, as well as unit values, subtotals, totals including VAT, and any applicable advance payments.
- **RN-M2-07:** Data is sent to Siigo exclusively through structured batch export files, generated manually by the secretary from approved work orders. Automatic data entry or real-time synchronization with the accounting software is strictly prohibited.

- **RN-M2-08:** The comprehensive 360-degree customer profile, grouped by Tax ID, consolidates contact information, commercial history, and private notes, which are only visible to authorized roles. Furthermore, it triggers a risk concentration alert when a customer exceeds an established purchasing threshold. VAT is always included in the displayed price.

Assumptions Dictating the Design

1. The automatic session timeout after fifteen minutes of inactivity is accepted to protect information on the shared computer.
2. Computers are turned off at night. Without an overnight operational window, batch processes must strictly run at the end of the workday.
3. The implementation of the WhatsApp channel alongside the guided form must be explicitly confirmed with management.
4. The brand and model of the fingerprint scanner, as well as the exact format of the physical work order, remain to be validated on-site.

CRC Cards

Module 1 (Security and Backups)

Role	
Responsibilities	**Collaborators**
• Define one of the three system roles (Super Administrator, Administrator, Operator). • Hold the set of permissions tied to the role, never to the person. • Answer whether the role grants a given permission.	• User

User	
Responsibilities	**Collaborators**
• Represent an individual employee account: full name, unique email and account status. • Hold exactly one mandatory role; change it when authorized. • Activate or deactivate the account and update the profile. • Check permissions through its role and record its last access.	• Role • TOTPEnrollment • Session • AuditEntry

TOTPEnrollment	
Responsibilities	**Collaborators**

• Store the secret that links the user with the authenticator app. • Know its expiration time (five minutes) and report whether it has expired. • Be confirmed or invalidated; a user has at most one.	• User

Session	
Responsibilities	**Collaborators**
• Represent a login session with its status and last activity time. • Update its last activity and close itself after 15 minutes of inactivity. • Be terminated when a new login replaces it (one active session per account). • Report whether it is still active.	• User

AuditEntry	
Responsibilities	**Collaborators**
• Record, in an unalterable way, who did a critical action, what it was, on which entity and when. • Be append-only: it can never be edited or deleted.	• User

BackupFile	
Responsibilities	**Collaborators**
• Represent one backup copy: type (daily, weekly or monthly), SHA-256 hash and integrity status. • Compare its stored hash with a computed one to verify integrity. • Mark the result of the integrity check.	• RestoreProcess

RestoreProcess	
Responsibilities	**Collaborators**
• Track each restoration attempt made from a backup. • Register the result of the test in the isolated sandbox before touching production. • Allow confirming and completing the restoration only if the verification passed.	• BackupFile • User

Module 2 (Commercial and Virtual Assistant)

Product	
Responsibilities	**Collaborators**
• Represent a catalog item: unique code, name, official price and status. • Change its price, only by management and with a support document. • Be activated or deactivated and tell whether it can be selected in a quotation or work order.	• PriceChangeAudit • QuotationItem • WorkOrderItem

PriceChangeAudit	
Responsibilities	**Collaborators**
• Keep an unalterable record of each price change: previous price, new price, support document, who and when.	• Product • User

Customer *«abstract»*	
Responsibilities	**Collaborators**
• Hold the data shared by every customer: tax ID, contact information and status. • Group the 360-degree profile: contact, commercial history and private notes. • Let subclasses define how the customer name is displayed and its type. • Be activated or deactivated.	• NaturalPerson • LegalEntity • CommercialNote • Quotation • WorkOrder

NaturalPerson *extends Customer*	
Responsibilities	**Collaborators**
• Represent an individual customer identified by their ID document number. • Provide the person’s full name as the display name.	• Customer

LegalEntity *extends Customer*	
Responsibilities	**Collaborators**
• Represent a company customer identified by its NIT. • Provide the business name as the display name.	• Customer

CommercialNote	
Responsibilities	**Collaborators**
• Store a private follow-up note about a customer with its content and author. • Be edited only by authorized roles. • Feed the customer’s 360-degree profile.	• Customer • User

Quotation	
Responsibilities	**Collaborators**
• Represent a quotation: customer (optional for potential customers), items, total amount and state. • Recalculate the total when items are added or removed. • Be approved, modified or rejected by the owner, always stating the reason when rejected. • Never advance in the workflow without human approval.	• QuotationItem • Customer • ProcessingQueueEntry • User

QuotationItem	
Responsibilities	**Collaborators**
• Represent one quotation line: product, quantity and the unit price at the time of quoting. • Calculate its subtotal and allow changing quantity and unit price.	• Quotation • Product

WorkOrder	
Responsibilities	**Collaborators**
• Represent a work order with a unique sequential number, customer, items and agreed delivery date. • Follow its cycle: pending, logged in Siigo, reconciled. • Add, update and remove items, reschedule the delivery date and be cancelled with a reason.	• WorkOrderItem • Customer • Product • User

WorkOrderItem	
Responsibilities	**Collaborators**
• Represent one work order line: product and quantity to manufacture. • Allow changing the quantity.	• WorkOrder • Product

ProcessingQueueEntry	
Responsibilities	**Collaborators**
• Represent an approved quotation waiting to be exported to Siigo. • Keep its export status and the time it was enqueued. • Be marked as exported by a batch; rejected quotations never enter the queue.	• Quotation • ExportBatch

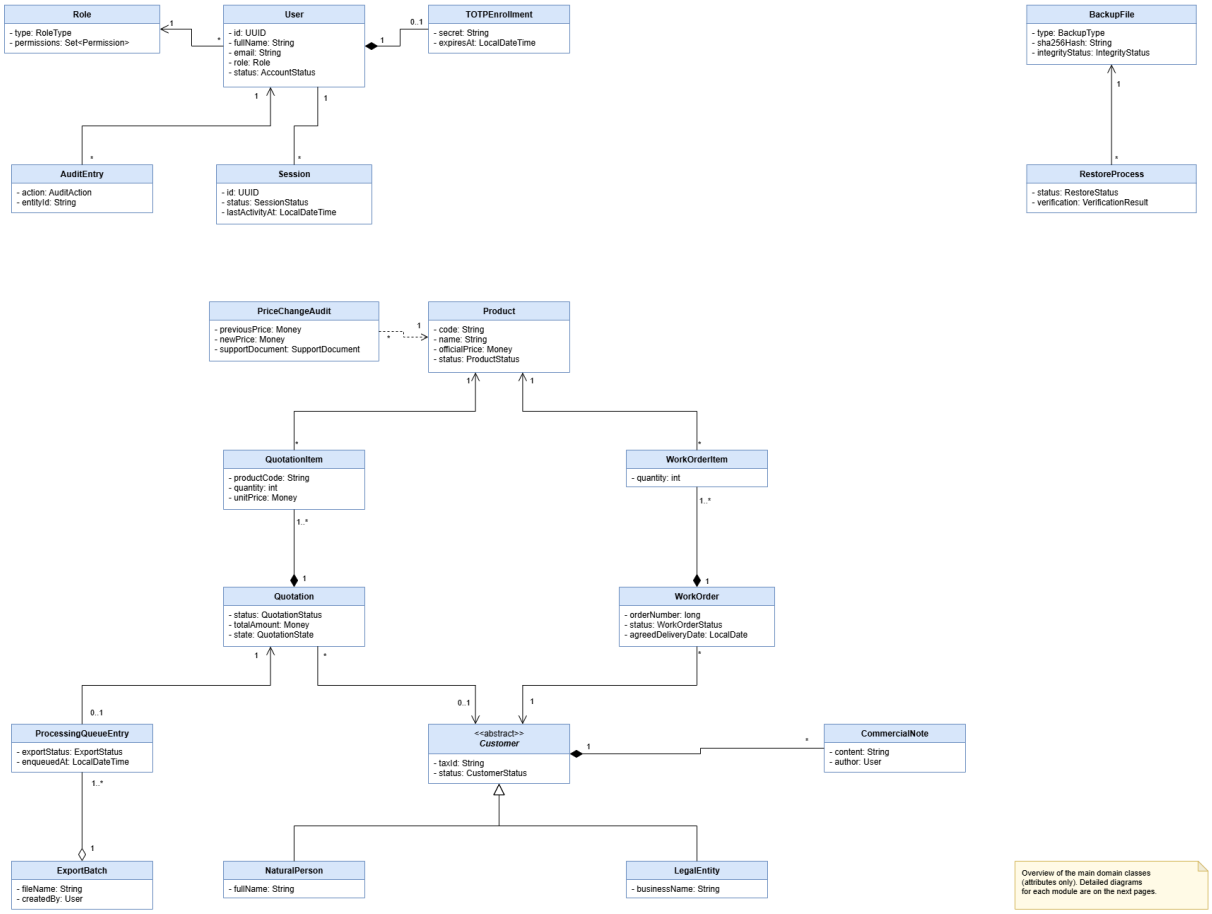
ExportBatch	
Responsibilities	**Collaborators**
• Represent the batch file generated manually by the secretary. • Group one or more queue entries into a single structured file to upload into Siigo. • Know who created it and how many orders it contains.	• ProcessingQueueEntry • User

Class Diagram

The UML class diagram was developed using draw.io and is structured into twelve sections, numbered from 0 to 11. These sections are organized as follows: 0 - Domain Overview, 1 - Authentication & Sessions, 2 - Users & RBAC, 3 - Backup Creation & Retention, 4 - Backup Inventory & Restore, 5 - Product Catalog & Price Audit, 6 - Customers & Commercial Notes, 7 - Quotations: Domain & Review, 8 - Chatbot & Quotation Request, 9 - Work Orders, 10 - Export Queue & Accounting, and 11 - Shared Kernel.

The primary section is 0 - Domain Overview, which provides a comprehensive global view of the domain. This section outlines the system's core entities -such as Role, User, TOTPEnrollment, Session, BackupFile, RestoreProcess, AuditEntry, Product, PriceChangeAudit, Customer, NaturalPerson, LegalEntity, CommercialNote, Quotation, QuotationItem, WorkOrder, WorkOrderItem, ProcessingQueueEntry, and ExportBatch- along with the structural relationships between them. The remaining eleven sections delve into the technical and operational specifics of each individual subdomain. An image representing this initial section is attached.

Domain Overview



To view and analyze the complete diagram, you can access it directly via this hyperlink. Additionally, the .drawio file is available in the project repository and simply needs to be opened in draw.io.

Relational Database Model

The relational model was designed for PostgreSQL and covers Modules 1 (Security and Backups) and 2 (Commercial and Virtual Assistant). It consists of seventeen tables, which are strictly necessary to support the business rules without requiring additional auxiliary tables.

Tables for Module 1 (Security and Backups)

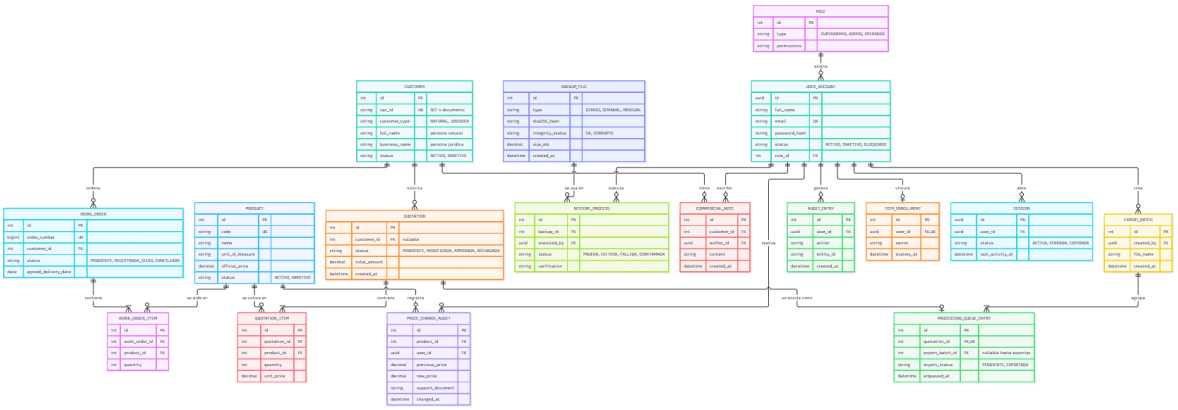
- **ROLE:** Defines the three system roles (Super Administrator, Administrator, and Operator) along with their respective permissions. Permissions are tied exclusively to the role and not to the individual.
- **USER_ACCOUNT:** Stores employee accounts, including the name, a unique email address, the encrypted password, account status, and assigned role.
- **TOTP_ENROLLMENT:** Manages the link with the authenticator app for two-factor authentication. It stores the cryptographic secret and its five-minute expiration time. Each user can have a maximum of one active enrollment.

- **SESSION:** Manages login sessions. It records the status and last activity to force an automatic logout after fifteen minutes of inactivity, while also ensuring that only one active session exists per account.
- **AUDIT_ENTRY:** Functions as the general audit log for critical actions (logins, role changes, backup executions). It records who performed the action, what they did, and when. This table is append-only; editing or deleting records is strictly prohibited.
- **BACKUP_FILE:** Acts as the backup inventory. It records the backup type (daily, weekly, or monthly), file size, creation date, and the SHA-256 hash to verify data integrity.
- **RESTORE_PROCESS:** Documents every data restoration attempt. It references the backup used, the user who executed the action, and logs the result of the test performed in the isolated sandbox environment before being applied to production.

Tables for Module 2 (Commercial and Virtual Assistant)

- **PRODUCT:** Represents the item catalog. It contains the unique code, name, unit of measure, official price, and product status.
- **PRICE_CHANGE_AUDIT:** Retains the price modification history. It logs the previous price, the new price, the responsible user, the date, and the justification document. This record is mandatory and cannot be altered.
- **CUSTOMER:** Manages customer information, identified by their Tax ID (NIT) or national ID. It uses a single record type to cover both individuals and corporate entities through the *customer_type* field.
- **COMMERCIAL_NOTE:** Stores private follow-up notes regarding a specific customer along with the note's author. This information feeds into the comprehensive 360-degree customer profile.
- **QUOTATION:** Contains the quotation header, including the customer (which is optional since potential customers do not need a registered account), document status, and total value.
- **QUOTATION_ITEM:** Details the line items for each quotation, specifying the product, requested quantity, and the unit price applicable at the time of quoting.
- **WORK_ORDER:** Represents the work order (W.O.). It includes a unique sequential number, the associated customer, the process status (pending, logged in Siigo, or reconciled), and the agreed delivery date.
- **WORK_ORDER_ITEM:** Breaks down the line items corresponding to each work order, indicating the product and the quantity to be manufactured.
- **PROCESSING_QUEUE_ENTRY:** Functions as the waiting queue for Siigo. Every approved quotation enters this table waiting to be exported.
- **EXPORT_BATCH:** Logs the export batch generated manually by the secretary. It groups multiple queue entries into a single structured file for subsequent upload into the Siigo accounting system.

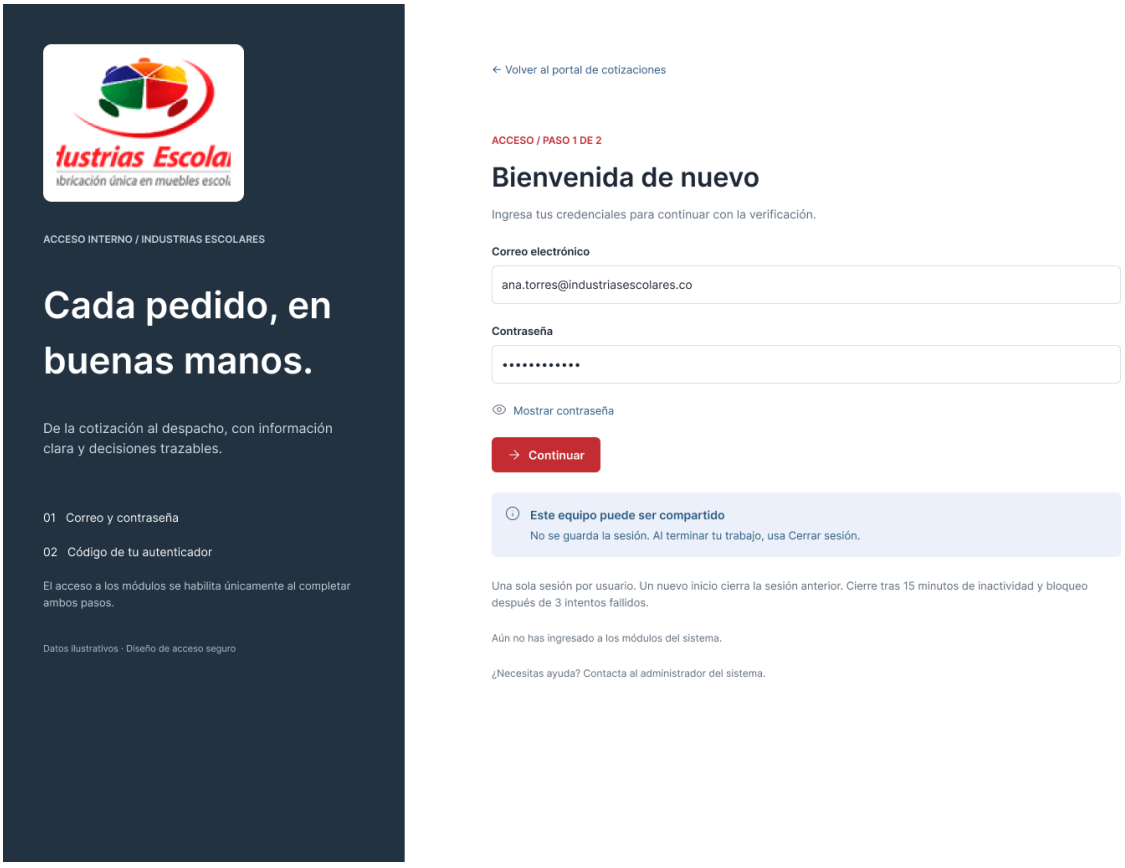
The complete entity-relationship diagram, along with its Mermaid source code, is attached in this same directory within the project repository. Furthermore, this diagram can be accessed via Mermaid using the provided hyperlink.



Mockups

The Industrias Escolares management system has three fixed user profiles: Super Administrator, who is Carlos Ríos; Administrator, who is Ana Torres; and Operator, who is Marta Ruiz. Since the equipment may be shared, all screens remind users to log out when they leave.

Login

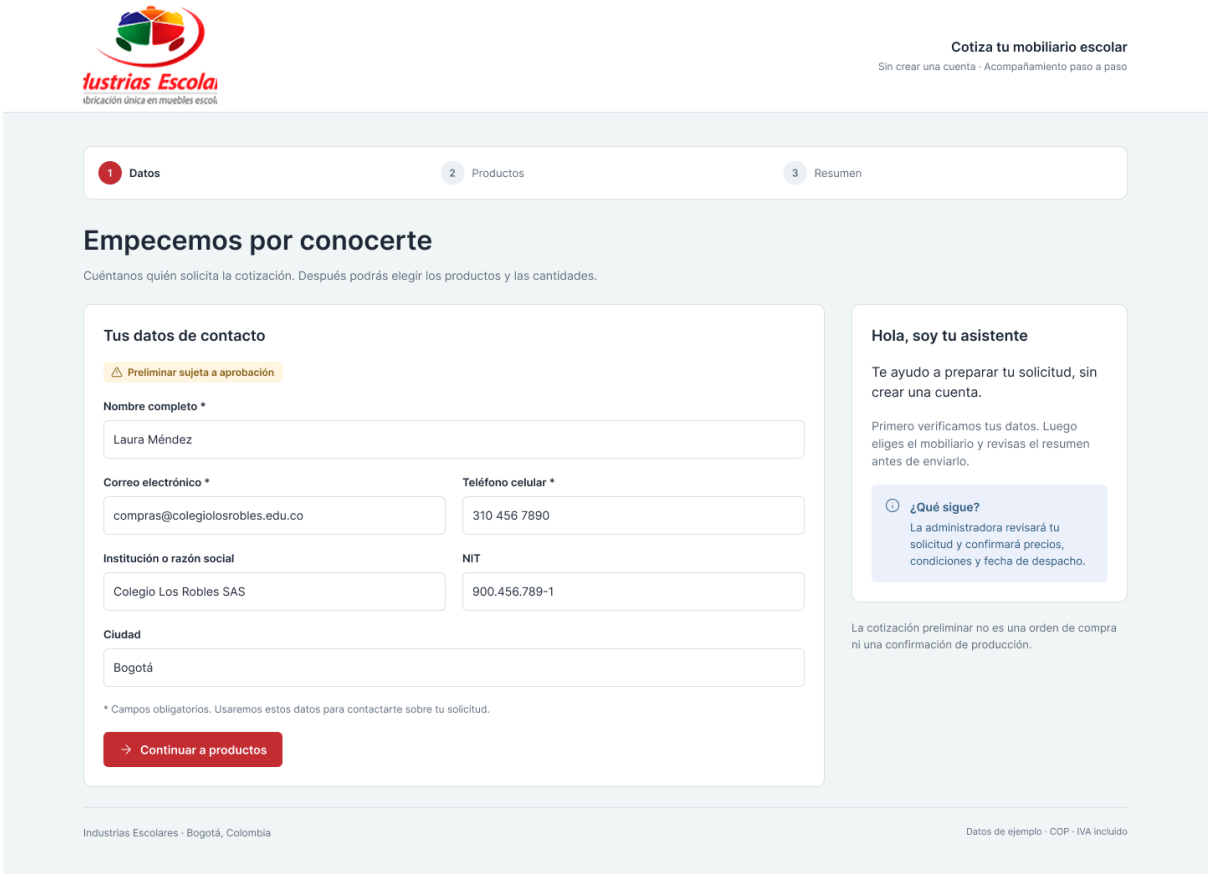


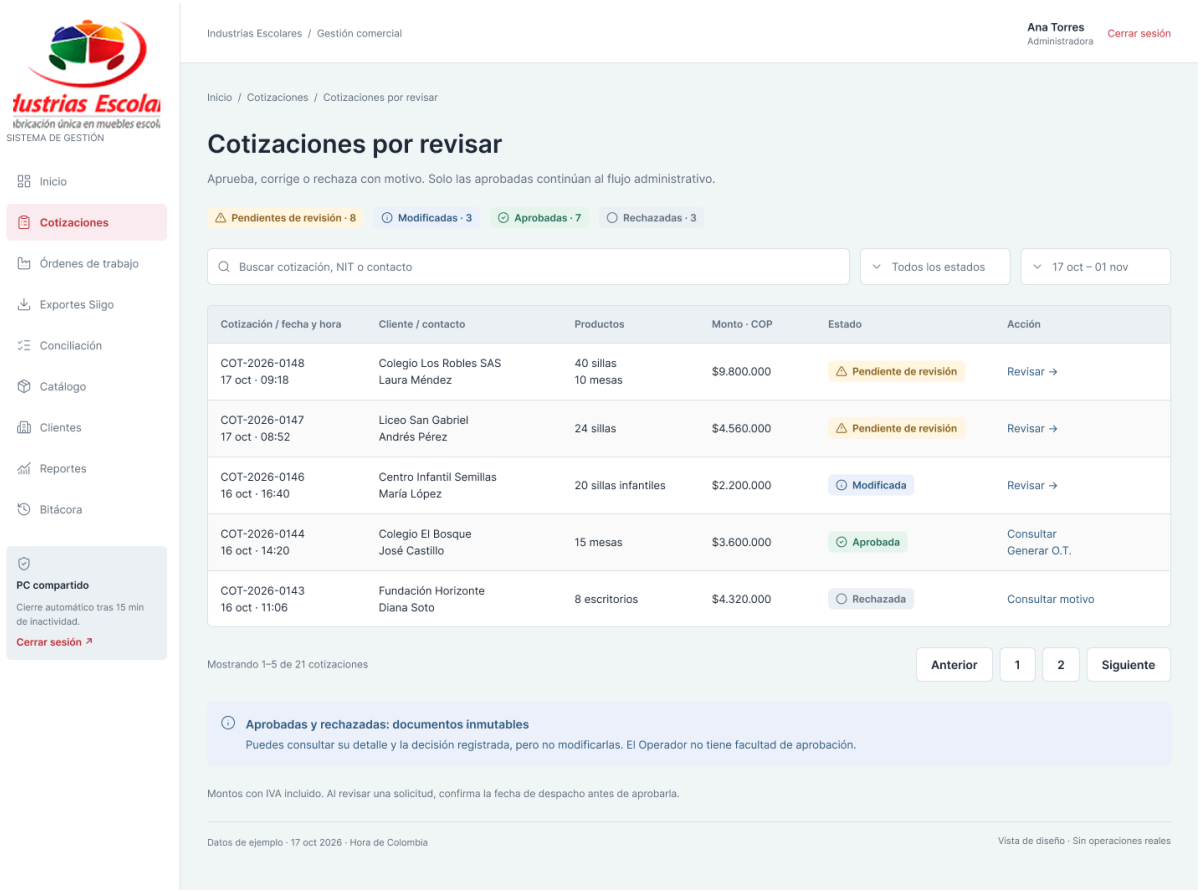
Internal access takes place in two steps. First, the user enters their email address and password. Then they open their authenticator app and enter the six-digit code, which is not

sent by email or text message. If the code is incorrect or has expired, they can wait for the next one and try again.

On the first login, an additional screen appears to link the authenticator by scanning a QR code or entering a key. If the linking code expires, a new one is generated. After three failed attempts, the account is locked, and the user must contact the administrator for assistance. Other scenarios are also covered: a session that expires after fifteen minutes of inactivity, a session that is closed because someone logged in from another device, and a notification indicating that the user's role does not have permission to perform an action.

Quotations

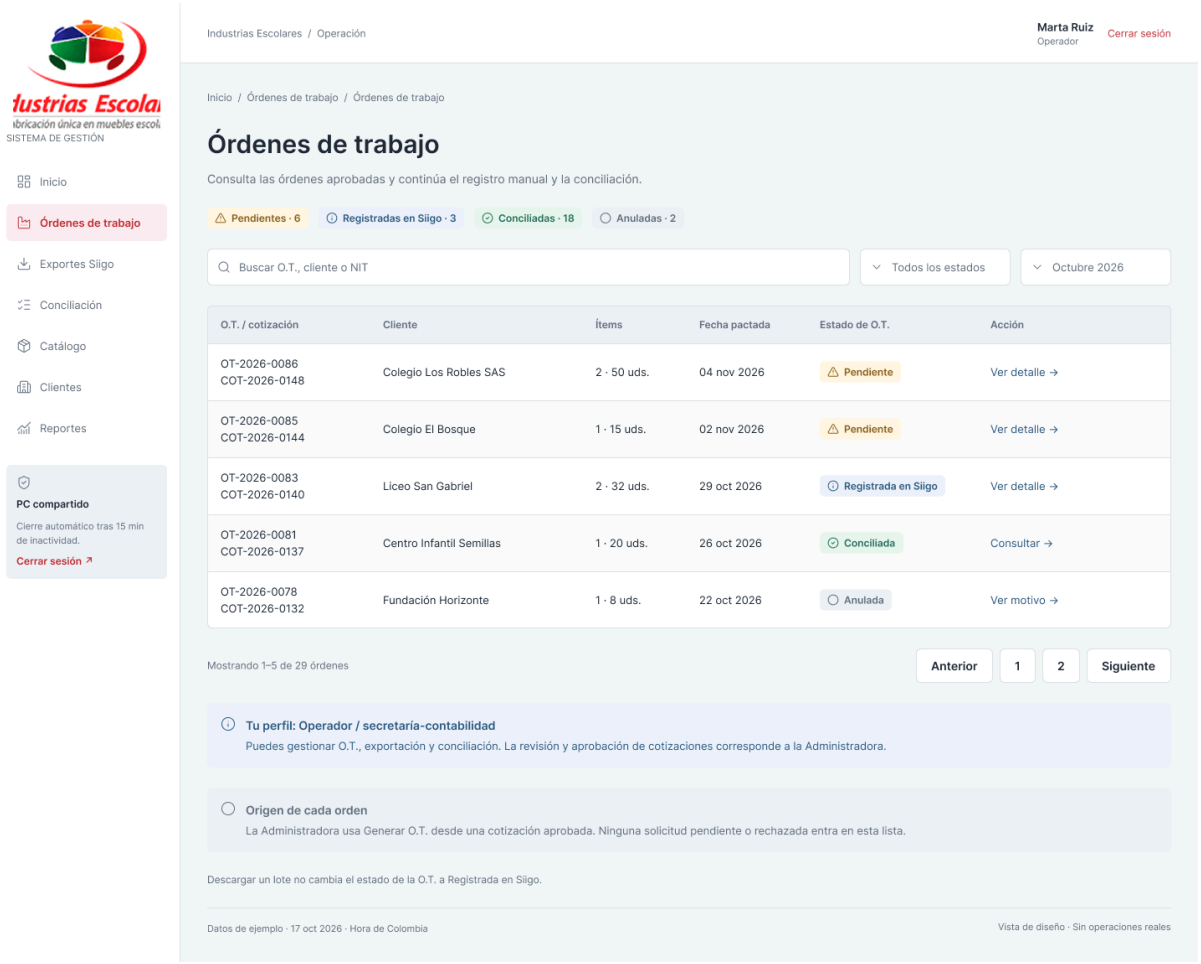




This section has two sides. Customers can proceed through three steps without creating an account. First, they provide their contact information and institution details. Then they select products and quantities, while a side panel displays the running total. Finally, they receive a preliminary quotation, which they can download as a PDF or receive by email. This quotation is subject to approval, so prices, terms, and the dispatch date are not yet final.

On the internal side, the Administrator has an inbox with requests that can be filtered by status: pending, modified, approved, and rejected. When opening a request, they can adjust quantities, prices, and commercial terms, always explaining the reason for the changes, and must confirm the dispatch date before approving it. There are three possible actions: approve, save a revision, or reject with a reason and an additional confirmation. Once approved or rejected, the quotation can no longer be modified. Only an approved quotation enables the creation of a work order.

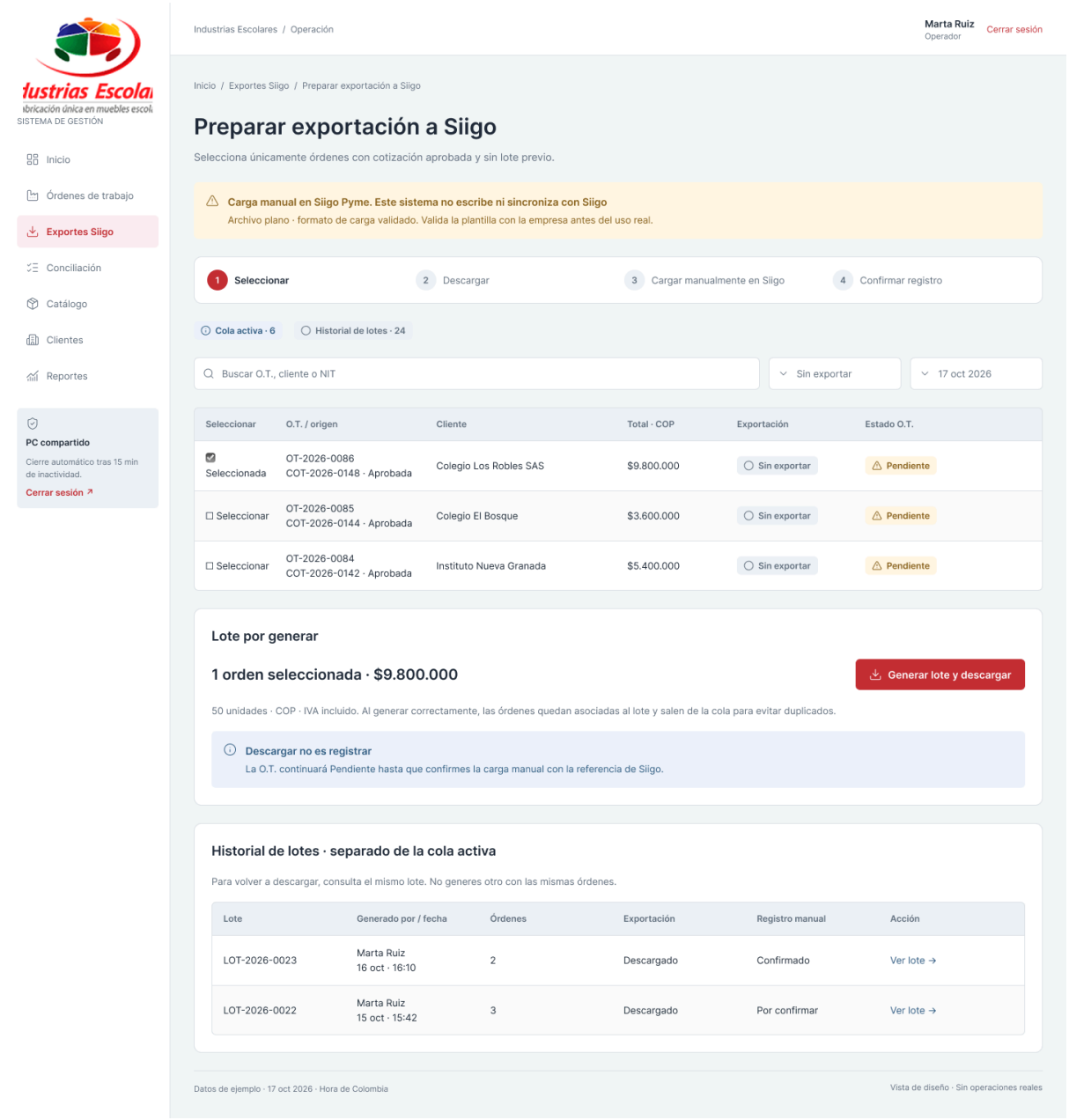
Work Orders



The Operator sees a list of all work orders, including the customer, the number of items, and the agreed date. Each order moves through three statuses in sequence: Pending, Registered in Siigo, and Reconciled. There is also a Cancelled status. Skipping statuses is not allowed. The details show the order's origin, customer information, products with their manufacturing specifications, the total amount, the thirty percent down payment, and the outstanding balance.

The Administrator generates the work order only once from an approved quotation, and the system prevents a second order from being created for the same quotation. While the order is Pending or Registered, it can be edited or cancelled, provided a reason is recorded. Once it is Reconciled or Cancelled, it becomes read-only.

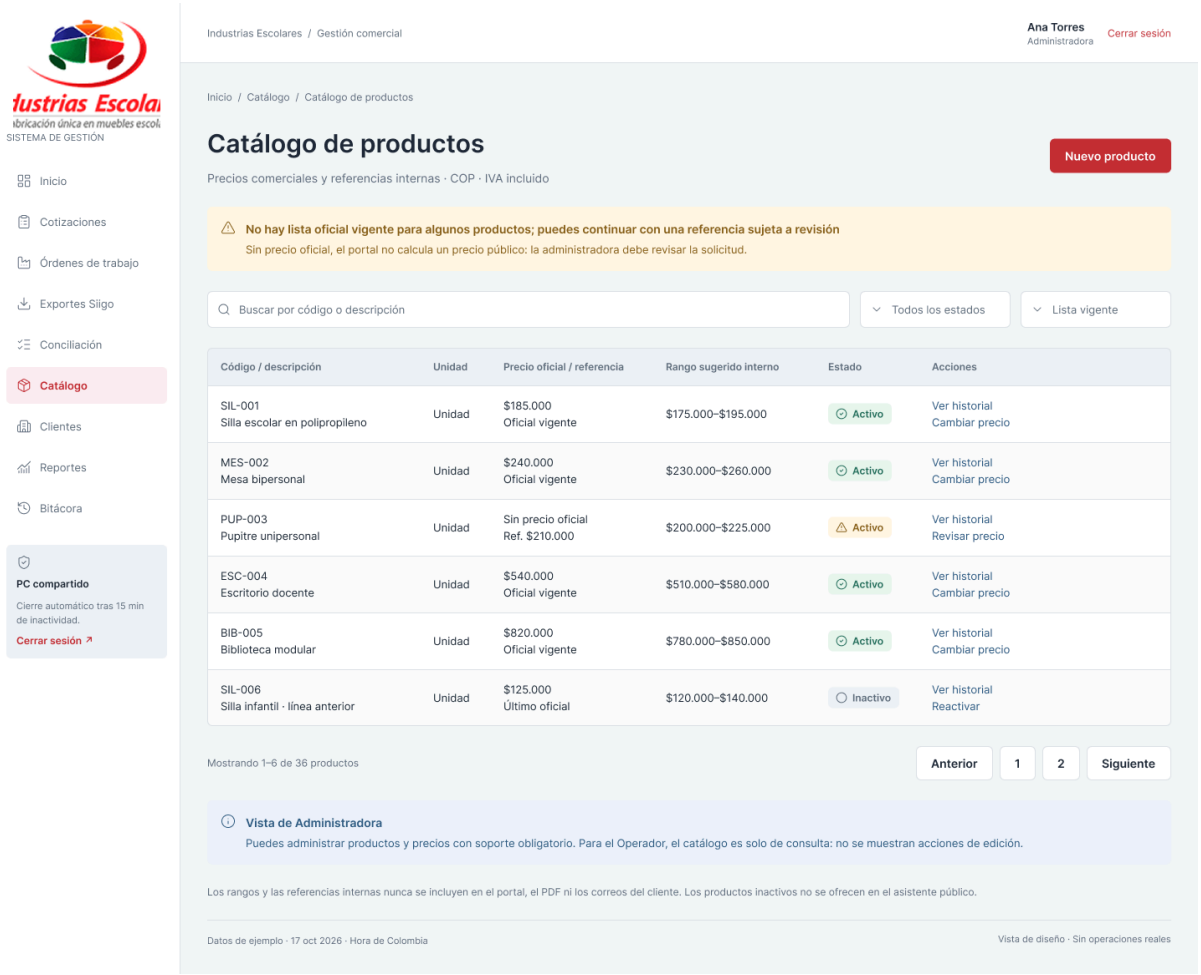
Exports to Siigo



This module does not connect to Siigo. All entries in Siigo are made manually, and the screen makes this clear from the outset. The Operator selects approved orders that have not yet been assigned to a batch, generates the batch, and downloads a flat file. They then upload the file to Siigo Pyme themselves. Once finished, they return to the system, enter the reference of the document created in Siigo, indicate that they have verified the amounts, and confirm the registration.

It is important to note that downloading the file does not change the order's status. Only the subsequent confirmation marks it as Registered. There is a batch history separate from the active queue, where a batch can be downloaded again without generating another batch containing the same orders. If file generation fails, no orders are marked or excluded, and the process can be retried.

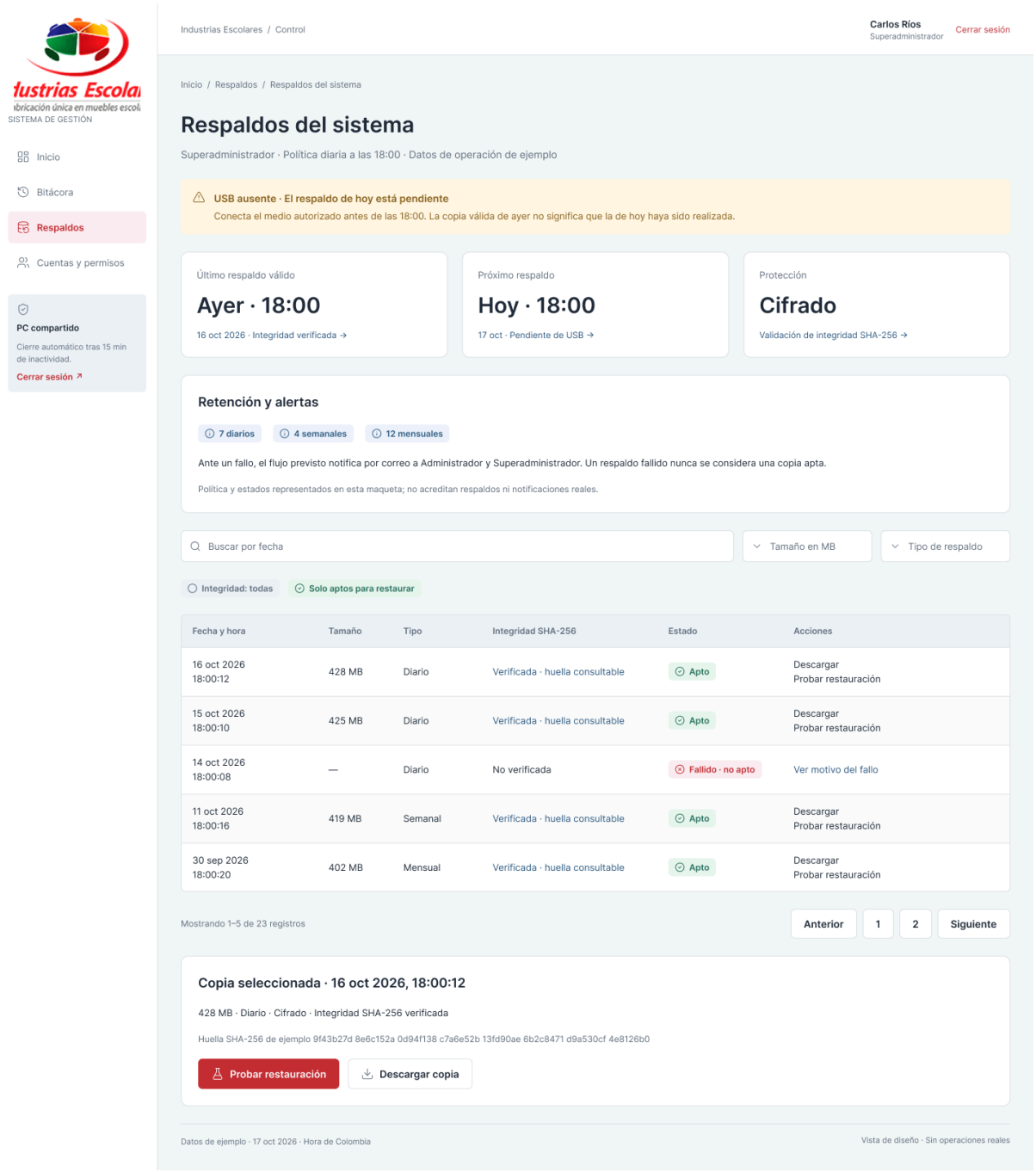
Catalog



The list displays each product's code, official or reference price, suggested internal price range, and status, with search and filtering options. If a product does not have a current official price list, the system warns that its price requires review and that the portal does not calculate a public price for it. Internal price ranges never appear in the portal, PDFs, or customer emails, and inactive products are not offered to the public.

Only the Administrator can change a price. To do so, they must provide a reason, attach a supporting document in PDF or image format, and confirm the change. The new price applies only to future transactions, so existing quotations and work orders retain their original amounts. Every change is recorded along with the date, the person responsible, and supporting evidence. The Operator can only view the catalog.

Backups



This section is exclusively for the Super Administrator. The dashboard displays the last valid backup, when the next one is due, that the backup is performed at six in the evening on a USB drive, and that it is encrypted. If the USB drive is not connected, an alert appears because having a valid backup from yesterday does not guarantee that today's backup has been completed. The retention policy keeps seven daily, four weekly, and twelve monthly backups. If a backup fails, an email notification is sent, and the backup is never considered valid. The table allows users to search and filter backups, check their integrity, download them, or test their restoration.

Restoring a backup is a four-step process. First, the backup is selected. Then, an isolated test is performed in a separate environment that does not affect production. The results are reviewed, and finally, the restoration is confirmed in production. For this final confirmation, the user must check an authorization box and type the word RESTORE. If the test fails, the restoration is cancelled, and the production environment remains untouched.

Other Views

The Administrator's home screen brings together the day's pending tasks: quotations to review, orders to register, and orders to reconcile. It includes a five-step summary of the entire process and a weekly report of requests.

Reconciliation compares a work order against the Siigo document line by line, including quantities, subtotals, and the total amount. To reconcile an order, the user must attach supporting evidence, enter a note, and confirm that they have completed the verification. The order then becomes read-only and cannot be moved back to a previous status. If there are discrepancies, the final action remains disabled until they are resolved.

The 360-degree customer profile allows users to search by tax identification number (NIT) or legal business name and view each customer's quotation and work order history. The NIT and entity type cannot be changed. Customers are not deleted; they are only deactivated, and their history is retained. The profile also includes private notes that never appear in PDFs or emails, as well as an alert when a customer accounts for a very large share of total sales.

The audit log is read-only for all user profiles. It records who did what and when, with filters for actor, action, entity, and date. Each event allows users to open its supporting evidence. Finally, under Accounts and Permissions, the Super Administrator can create, edit, and deactivate accounts using the three fixed profiles, assign an initial password, and allow users to link their authenticator on their first login.

Overall Workflow

Everything begins when a customer requests a quotation through the portal. The Administrator reviews it, makes corrections if necessary, and approves it, enabling the creation of a work order. The Operator includes the order in a batch, uploads it manually to Siigo, and confirms its registration using the document reference. Finally, the order is reconciled against Siigo and closed as a read-only record. Every step leaves an audit trail, while the Super Administrator is responsible for backups and account management.

Note: *To view all the designs for the additional windows, as well as those for the other views, you can visit the following link or this other one.*

Declaration of the Use of Artificial Intelligence

For the preparation of this document, we declare that the use of Artificial Intelligence tools—specifically Gemini and Claude—was strictly limited to support, review, and translation purposes. First, these tools were used exclusively to translate the system

domain, business rules, and technical descriptions into English. Likewise, we relied on them to conduct brainstorming sessions to explore new proposals and refine our design components, a process that was carried out only after we had independently defined and structured the system's core functionalities, database models, and mockups.

Furthermore, Gemini and Claude were employed as a validation mechanism to ensure there were no weak points or gaps in the project's overall technical reasoning. This included reviewing the consistency between the relational database model, the UML class diagrams, the business rules, and the proposed mockup workflows to guarantee their logical soundness and architectural integrity. Finally, we used both artificial intelligences to confirm that each of these design elements strictly complied with the correct structure and formal presentation required for this academic submission.